Reg. No. | | | | | | | | | |

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER B.TECH. (INSTRUMENTATION AND CONTROL ENGG.)

END SEMESTER DEGREE EXAMINATIONS, MARCH - 2021

SUBJECT: Digital Image Processing [ICE 4021]

TIME: 3 HOURS 19-03-2021 MAX. MARKS: 50

Instructions to candidates :*Answer ALL questions and missing data may be suitably assumed.*

1A. With an example for each, explain the process of sampling and quantization.

1B. Define following terms: i) Pixel ii) Image negative

1C. What are the various processing levels in digital image processing? Give an example for each.

(4+2+4)

2A. For the given kernel (w) and image (f), compute the convolution of two.

$$w = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \qquad f = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2B. Discuss the limiting effect of repeatedly subtracting Image 2 from Image 1. Assume that the results have to be represented in 8 bits.

2C. What is histogram matching? With an example compare histogram equalization and histogram specification.

(2+3+5)

3A. With the mathematical expressions explain harmonic and contra harmonic filter. Which type of degradation can be removed using these two techniques?

3B. Explain i) Sobel operators ii) Edge enhancement

3C. With an example, discuss averaging spatial filter with a box and Gaussian kernels.

(3+3+4)

4A. Compare subjective and objective fidelity criteria with a suitable example.

4B. What is entropy? Discuss its significance in medical image analysis.

4C. What are lossless compression techniques? Compute the Huffman coding for the following probability distribution. A = {0.4, 0.3, 0.1, 0.1, 0.06, 0.04}

(3+3+4)

5A. Explain detection of shape of an object in outdoor environment.

5B. Write about performing image matching using DFT. Illustrate with suitable example.

5C. Discuss advanced healthcare system which employs image processing algorithms.

(3+3+4)